Lesson 2 - Inverting Matrices

2.1 Multiplication of matrices

Question 16

Level of difficulty: E

Given the three matrices A = $\begin{pmatrix} 0 & 1 & 2 \\ -2 & 2 & 3 \end{pmatrix}$, B = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, and C = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \\ 0 & 1 \end{pmatrix}$, check whether the following arithmetic operations are defined. If so, calculate.

a) A · B b) B · C

Question 17

Level of difficulty: E

Given the two matrices A = $\begin{pmatrix} 0 & 1 \\ -2 & 2 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}$, calculate A · B. Justify why the arithmetic operation is possible.

Solution question 16

Solution hint: Look at the types of matrices.

Short solution: a) possible: C = $\begin{pmatrix} 0 & 5 \\ -4 & 7 \end{pmatrix}$ b) not possible

Detailed solution: A of type 2x3, B of type 3x2, C of type 3x2, number of columns of the first matrix must be equal to the number of rows of the second matrix. Each row is multiplied by each column.

a) possible: C = $\begin{pmatrix} 0 & 5 \\ -4 & 7 \end{pmatrix}$ b) not possible

Solution question 17

Solution hint: Row times column respectively, consider the types of matrices.

Short solution: The number of columns of A matches the number of rows of B:

C = $\begin{pmatrix} 0 & 2 \\ -2 & -2 \end{pmatrix}$

Detailed solution: A has two columns, B has two rows. The number is the same. Multiply the rows by the columns: C = $\begin{pmatrix} 0 & 2 \\ -2 & -2 \end{pmatrix}$

Question 18

Level of difficulty: M

Calculate the dyadic product of the two vectors $\vec{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\vec{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$.

Question 19

Level of difficulty: M

Given two lower triangular matrices A = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, show that C = A · B is also a lower triangular matrix.

Solution question 18

Solution hint: Dyadic product $\vec{a} \cdot \vec{b}^{\mathrm{T}}$, a matrix is formed.

Short solution: $\vec{a} \cdot \vec{b}^{\mathrm{T}} = \begin{pmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 9 & 6 & 3 \end{pmatrix}$

Detailed Solution: $\vec{a} \cdot \vec{b}^{\mathrm{T}} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot (3\ 2\ 1) = \begin{pmatrix} 3 & 2 & 1 \\ 6 & 4 & 2 \\ 9 & 6 & 3 \end{pmatrix}$

Solution question 19

Solution hint: Calculate A · B.

Short solution: C = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$

Detailed solution: C = A · B = $\begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$. The resulting matrix is also a lower triangular matrix.

Question 20

Level of difficulty: D

Using the two matrices A = $\begin{pmatrix} 2 & 1 \\ 0 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 0 & 1 \\ 3 & 1 \end{pmatrix}$, show that $B^T \cdot A^T = (A \cdot B)^T$ holds.

Solution question 20

Solution hint: Calculate the left and right sides.

Short solution: $B^T \cdot A^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$, $(A \cdot B)^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$

Detailed solution: $B^T = \begin{pmatrix} 0 & 3 \\ 1 & 1 \end{pmatrix}$, $A^T = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}$, $B^T \cdot A^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$, $(A \cdot B)^T = \begin{pmatrix} 3 & 3 \\ 9 & 3 \end{pmatrix}^T = \begin{pmatrix} 3 & 9 \\ 3 & 3 \end{pmatrix}$,
i.e., $B^T \cdot A^T = (A \cdot B)^T$

2.2 Properties of matrix multiplication

Question 21

Level of difficulty: E

Given two matrices A = $\begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, calculate C = A · B. What conclusion can be drawn from this calculation?

Question 22

Level of difficulty: E

Given two diagonal matrices A = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix}$, what is C = A · B? What can you conclude from this?

Solution question 21

Solution hint: Which special form does matrix B represent?

Short solution: If a matrix is multiplied by a unit matrix, the matrix does not change. C = A

Detailed solution: C = A · B = $\begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 5 & 2 \\ -3 & 4 & 0 \\ 1 & 2 & 1 \end{pmatrix}$. C = A · E = A. If a matrix is multiplied by an identity matrix, it remains the same.

Solution question 22

Solution hint: Multiply the two matrices.

Short solution: C = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 12 \end{pmatrix}$. The result is a diagonal matrix.

Detailed solution: C = A · B = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 12 \end{pmatrix}$. The diagonal elements are multiplied. The result is a diagonal matrix.

Question 23

Level of difficulty: M

Does the commutative law apply to matrix multiplication? Explain your statement with the help of an example.

Question 24

Level of difficulty: M

The matrix A = $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 4 & 2 & 1 \end{pmatrix}$ is multiplied by a zero matrix. What type must this zero matrix be? Give an example and the result. What can you conclude from this?

Solution question 23

Solution: The commutative law is A · B = B · A.

Short solution: The commutative law does not apply.

Detailed solution: Choose two matrices A = $\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}$ and B = $\begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix}$.

A · B = $\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 4 & 10 \\ 5 & 14 \end{pmatrix}$, B · A = $\begin{pmatrix} 1 & 4 \\ 2 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & 5 \\ 10 & 4 \end{pmatrix}$, i.e., A · B ≠ B · A.

The commutative law does not apply.

Solution question 24

Solution hint: First, consider what type the zero matrix must be. Then perform the matrix multiplication.

Short solution: The zero matrix must have three rows. The result is a zero matrix that has as many rows as A and as many columns as the zero matrix.

Detailed solution: Since matrix A is of type 3x3, the zero matrix must have three rows; the number of columns does not matter.

Example: 0 = $\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$. A · O = $\begin{pmatrix} 2 & 3 & 1 \\ 1 & 4 & 2 \\ 4 & 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$

The result is a zero matrix that has as many rows as A and as many columns as 0.

Question 25

Level of difficulty: D

Two matrices A = $\begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ x & x & 0 \end{pmatrix}$ and B = $\begin{pmatrix} 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{pmatrix}$, each with placeholders *x*, are multiplied. At which points can C = A · B assume values not equal to zero?

Solution question 25

Solution hint: Perform the matrix multiplication.

Short solution: The following elements of C are not equal to zero: c_{22}, c_{23}, c_{32}, c_{33}

Detailed solution: C = A · B = $\begin{pmatrix} 0 & 0 & 0 \\ x & 0 & 0 \\ x & x & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & x & x \\ 0 & 0 & x \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & y & y \\ 0 & y & y \end{pmatrix}$. Only the following elements of the matrix C can be non-zero: c_{22}, c_{23}, c_{32}, c_{33}.

2.3 Inverse matrices

Question 26

Level of difficulty: E

Check whether the inverse of the following matrices can be formed.

a) A = $\begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}$ b) B = $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$

Question 27

Level of difficulty: E

Is the matrix A = $\begin{pmatrix} 2 & 1 \\ 3 & 4 \end{pmatrix}$ invertible? Justify your answer.

Solution question 26

Solution hint: Consider which conditions must be fulfilled for a matrix to be invertible.

Short solution: a) no b) no

Detailed solution: a) No, because A is a square diagonal matrix. However, not all diagonal values are non-zero. b) No, because only square matrices are invertible.

Solution question 27

Solution hint: Which condition for a 2x2 square matrix must be met for the inverse to exist?

Short solution: det not equal to zero -> invertible

Detailed solution: det = ad – bc = 8 – 3 = 5 $\neq$ 0. Since the determinant is not zero, the inverse exists.

Question 28

Level of difficulty: M

Determine the inverse of the matrix A = $\begin{pmatrix} -2 & 3 \\ 1 & 4 \end{pmatrix}$.

Question 29

Level of difficulty: M

Determine the inverse of the matrix D = $\begin{pmatrix} 2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{pmatrix}$

Solution question 28

Solution hint: Remember the formula that represents how to determine the inverse of a 2x2 square matrix.

Short solution: $A^{-1} = \begin{pmatrix} -\frac{4}{11} & \frac{3}{11} \\ \frac{1}{11} & \frac{2}{11} \end{pmatrix}$

Detailed solution: A = $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $A^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{-8-3}\begin{pmatrix} 4 & -3 \\ -1 & -2 \end{pmatrix} = \begin{pmatrix} -\frac{4}{11} & \frac{3}{11} \\ \frac{1}{11} & \frac{2}{11} \end{pmatrix}$

Solution question 29

Solution hint: Remember the formula that represents how to invert a square diagonal matrix.

Short solution: $D^{-1} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$

Detailed solution: A square matrix is invertible if its diagonal elements are not equal to zero. The inverse is determined by finding the reciprocal of the diagonal elements.

D = $\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$, $D^{-1} = \begin{pmatrix} 1/a & 0 & 0 \\ 0 & 1/b & 0 \\ 0 & 0 & 1/c \end{pmatrix} = \begin{pmatrix} 1/2 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/4 \end{pmatrix}$

Question 30

Level of difficulty: D

Show by example that if A^{-1} is the inverse of A, then A is the inverse of A^{-1}.

Solution question 30

Solution hint: Choose a simple square matrix and show that $A \cdot A^{-1} = A^{-1} \cdot A$ holds.

Short solution: $A \cdot A^{-1} = A^{-1} \cdot A = E$

Detailed solution: Example: $A = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix}$, $A^{-1} = ½ \begin{pmatrix} 4 & -2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix}$,

$A \cdot A^{-1} = \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} \cdot \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $A^{-1} \cdot A = \begin{pmatrix} 2 & -1 \\ -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \cdot \begin{pmatrix} 1 & 2 \\ 1 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$,

i.e., $A \cdot A^{-1} = A^{-1} \cdot A = E$